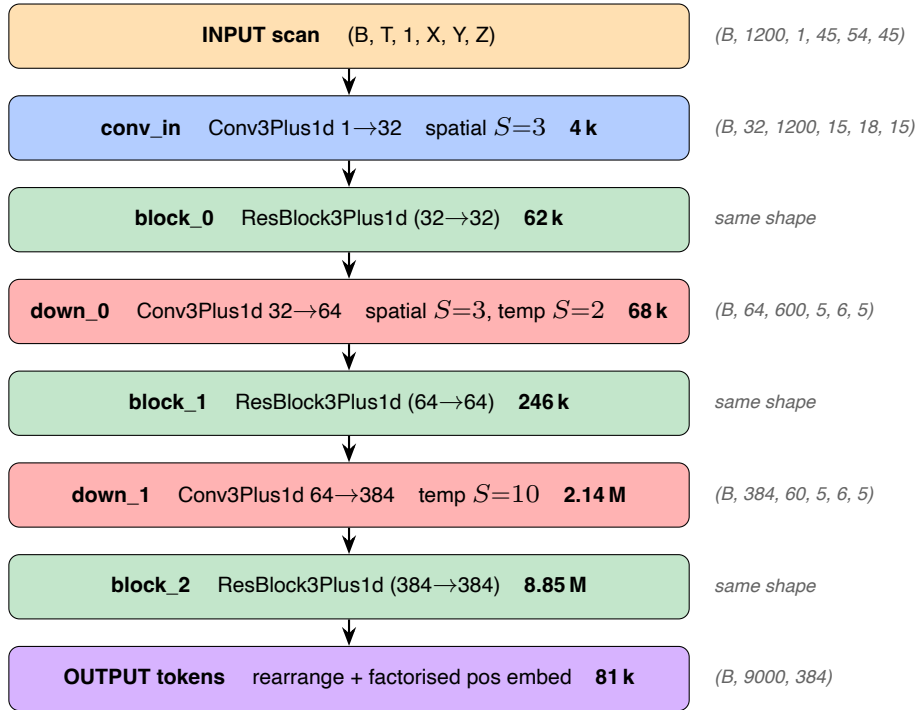


Patchify architecture — PatchEmbed3DPlus1D

2026-06-14

Hierarchical 3D+1D patchify encoder, inspired by MovieGen TAE (github.com/MathieuTuli/MovieGen, `tae.py:740`).
Basic block = Conv3Plus1d = Conv3d spatial (per frame) + Conv1d temporal (per voxel-position) — separable substitute for the missing Conv4d.



Total patchify parameters : 11.45 M

Setup HCP-YA — $T=1200$, $\text{temporal_kernel}=20$, image $45 \times 54 \times 45$, $\text{patch_size}=9$.

Strides totaux : spatial $3 \cdot 3 \cdot 1 = 9$ (= patch_size), temporal $1 \cdot 2 \cdot 10 = 20$ (= temporal_kernel).

Grille de tokens : $T_{\text{eff}} = 60$, $N_{\text{spatial}} = 5 \cdot 6 \cdot 5 = 150$ □ **9 000 tokens** dim 384.

Pos embed factorisé : $\text{pos_temporal}(60 \times 384) + \text{pos_spatial}(150 \times 384) + \text{pos_cls}(1 \times 384) = 81 \text{ k params}$, vs 3.46 M pour une table plate (43x moins).

Fichier source : `dinov2/layers/patch_embed_3d_plus_1d.py`.